

Week 12 – Final Report Analysis

Predicting Healthcare Expenditure Risk in the US Non-Cancer Adult Population

**A Risk Adjustment Model Incorporating Demographic, Clinical,
Economy, and Healthcare Utilization across MEPS Panels 11–15
(Years 2006 – 2010)**



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1. Abstract:

This study develops a longitudinal healthcare expenditure risk framework using the Medical Expenditure Panel Survey (MEPS) Panels 11–15 (2006–2011). The objective is to construct an analytically rigorous dataset for predicting inflation-adjusted Year-2 healthcare expenditures among adult, non-cancer patients using Year-1 demographic, socioeconomic, clinical, and behavioral determinants. Following extensive data cleaning, cancer exclusion harmonization across panels, feature engineering, and exploratory data analysis (EDA), unsupervised patient segmentation was conducted using K-means clustering. Expenditures were normalized to constant 2011 dollars using Consumer Price Index (CPI) adjustments and transformed into quartile-based expenditure ranges to enable risk-stratified classification modeling. Results reveal strong socioeconomic gradients in healthcare utilization, age-based multimorbidity clustering, nonlinear BMI-cost relationships, payer mix transitions across the lifespan, and distinct patient phenotypes. The prepared dataset establishes a foundation for predictive modeling of high-cost risk in longitudinal healthcare data.

Keywords: Medical Expenditure Panel Survey, longitudinal analysis, healthcare utilization, chronic conditions, data cleaning, feature engineering, Patient Clustering.

2. Introduction:

Healthcare expenditure prediction is central to risk adjustment, resource allocation, and population health management. The MEPS longitudinal dataset provides a nationally representative framework to examine determinants of healthcare utilization and cost over time. This project focuses on predicting Year-2 healthcare expenditures using Year-1 characteristics, thereby supporting prospective risk stratification rather than retrospective cost explanation.

The study integrates demographic, socioeconomic, health status, lifestyle, and chronic disease burden indicators to build a structured analytic dataset suitable for advanced modeling and patient segmentation.

2.1 Objectives

The primary objectives of this analysis are:

- To consolidate and clean longitudinal MEPS data from five consecutive panels (11-15)
- To do meaningful features engineering.
- To conduct EDA for identifying patterns in healthcare utilization that drive total healthcare expense.
- To prepare a robust dataset for subsequent predictive modeling and policy analysis.

2.2 Data Overview

The dataset comprises 78,022 observations across 56 variables extracted from MEPS Panels 11, 12, 13, 14, and 15, corresponding to data collection years 2006 through 2010.

Category	Variables
Indicators	DUPERSID, PANEL, BEGRFY1
Demographic	AGEY1X, SEX, RACETHNX, MARRYY1X, REGIONY1
Health Factors/Lifestyle	MNHLTH1, RTHLTH1, JTPAIN1, VISION2, HEARNG2, PHYACT3, BMINDX5, ADSMOK2
Cancer Flag	CANCERY1, CANCERY2, ICD9CODX
Chronic Conditions	ANGIDXY1, ARTHDXY1, ASTHDXY1, BPMLDXY1, CHDDXY1, CHOLDXY1, DIABDXY1, EMPHDXY1, HIBPDXY1, MIDXY1, OHRTDXY1, STRKDXY1

Socioeconomic	POVCATY1, EMPST1
Insurance	INSCOVY1, MCDEVY1, MCREVY1, TRIEVY1, TOTSLFY1, TOTMCDY1, TOTMCRY1, TOTPRVY1, TOTTRIY1, TOTOPUY1
Healthcare Utilization	TOTEXPY1, TOTEXPY2, OBDRVY1, OPTOTVY1, IPDISY1, RXTOTY1, ERTOTY1, OBTOTVY1 OBDEXPY1, OPTEXPY1, IPTEXPY1, RXEXPY1, ERTEXPY1, OBTEXPY1

3. Literature review

Healthcare expenditure prediction has become increasingly important in health economics and policy due to the highly skewed nature of medical costs, where a small proportion of individuals account for a disproportionately large share of total spending. Prior studies using the Medical Expenditure Panel Survey (MEPS) have consistently shown that healthcare expenditures are influenced by demographic characteristics, chronic disease burden, and prior healthcare utilization, making predictive modeling both complex and essential.

Wang and Shi (2020) applied a Random Forest model to MEPS data (2000–2015) to predict medical expenditures among individuals diagnosed with diabetes. Their findings demonstrated that machine learning models outperform traditional statistical approaches in capturing nonlinear relationships in healthcare data. The study identified prior health conditions, demographic variables, and utilization measures as key determinants of expenditure, supporting the use of tree-based models in healthcare cost prediction.

Bloniarz et al. (2022) proposed a two-stage super learner approach for modeling healthcare expenditures, emphasizing the challenges posed by zero-inflated and highly right-skewed cost distributions. Their results suggest that hybrid modeling frameworks can better capture the structural complexity of healthcare spending compared to single-model approaches. This is particularly relevant for studies dealing with imbalanced expenditure categories, where high-cost individuals represent a minority but are of significant interest.

McClellan et al. (2023) explored the application of machine learning algorithms to improve imputation in MEPS data. Their study highlighted that preprocessing decisions, including handling missing values and survey-specific coding structures, significantly impact downstream predictive performance. This reinforces the importance of rigorous data cleaning and feature engineering in MEPS-based analyses, as implemented in the present study.

Petrović et al. (2025) examined the application of interpretable machine learning models in healthcare outcome prediction, specifically in acute ischemic stroke care. Although focused on clinical outcomes rather than expenditure, the study underscores the growing importance of model interpretability and real-world applicability in healthcare analytics. It emphasizes that predictive models should not only achieve strong performance but also provide actionable insights for decision-making.

Collectively, these studies establish that healthcare expenditure prediction requires advanced modeling techniques, careful handling of skewed and imbalanced data, and robust preprocessing pipelines. Building on this literature, the present study develops a longitudinal predictive framework using MEPS Panels 11–15 to classify Year-2 healthcare expenditure into risk categories. Unlike prior work that focuses primarily on regression-based cost estimation or disease-specific populations, this study emphasizes multiclass risk stratification and prioritizes the identification of high-cost individuals. By incorporating class-weighted modeling and patient segmentation, this research contributes to improving the practical relevance of healthcare expenditure prediction for policy and resource allocation.

4. Data Cleaning and Preprocessing:

4.1 Data Integration:

- A custom Python parser was developed to extract fixed-width fields using SAS layout specifications.
- Merged multiple MEPS longitudinal panels (11–15) using DUPERSID.

4.2 Data Filtering:

- Adults (AGEY1X $\geq$ 18)
- Non-cancer patients (identified and removed cancer cases using ICD-9 codes for Panel 11)

Panel Number	Number of records (before filtering)	Number of records (after filtering)
11	16533	10991
12	12440	7795
13	18287	11421
14	16221	10336
15	14541	9277
Total Records	78,022	49,820

4.3 Missing Value Handling

- Converted MEPS negative codes [-1 (Inapplicable), -7 (Refused), -8 (Don't Know), -9 (Not ascertained)] to missing values (NA).
- For Panel 11, Year-1 chronic condition information was replaced using Round 5 data.
- Dropped columns with more than 50% missing values.
- Imputed missing values:
 - Median for continuous variables (e.g., TOTEXPY2, BMINDX5)
 - Mode or default category for categorical variables (e.g., ADSMOK2, PHYACT3, MARRYY1X, VISION2, HEARNG2, JTPAIN1)
- Filled health status variables (RTHLTH1, MNHLTH1) with neutral category 3 ("Good").
- Filled socioeconomic variable EMPST1 with category 4 ("Not Employed") where missing.
- Handled missing year information using panel-year mapping.

4.4 Feature Engineering

- Created total chronic condition count (sum of binary disease indicators).
- Derived BMI categories (Underweight, Normal, Overweight, Obese).
- Generated age groups (18–29, 30–44, 45–59, 60–74, 75+).
- Applied inflation adjustment to expenditure variables
- ('TOTEXPY2', 'TOTEXPY1', 'TOTSIFY1', 'TOTMCDY1', 'TOTMCRY1', 'TOTPRVY1', 'TOTTRIY1', 'TOTOPUY1')
- to convert values into constant 2011 dollars using CPI.
- Built Combined Health Score using physical and mental health ratings.
- Applied log transformation to the target variable (TOTEXPY2) and created categorical target using quantile-based binning (Low, Medium, High expenditure groups).
- Converted selected categorical variables into binary indicators
- ('EMPST1', 'ADSMOK2', 'PHYACT3', 'JTPAIN1', 'VISION2', 'HEARNG2', 'TRIEVY1', 'MCREVY1', 'MCDEVY1').
- Applied one-hot encoding to nominal variables
- ('SEX', 'RACETHNX', 'MARRYY1X', 'REGIONY1', 'INSCOVY1').

5 Feature Engineering:

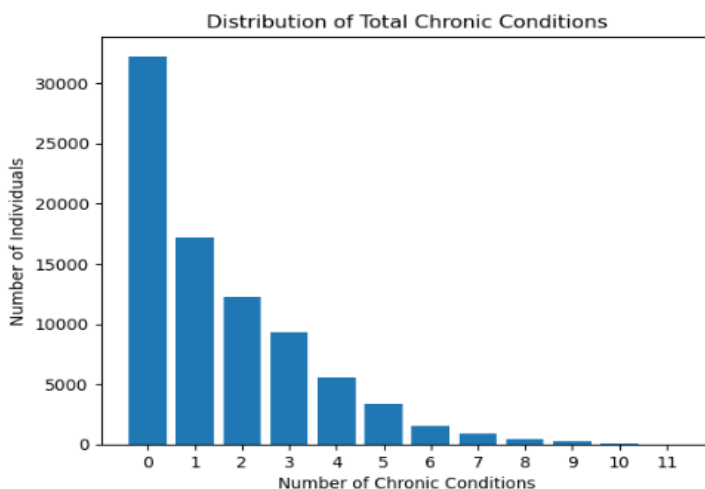
5.1 Chronic Disease Burden Index

To quantify comorbidity burden, a Total Chronic Conditions count variable was created by summing binary indicators across 11 chronic conditions:

Angina, Arthritis, Asthma, CHD, Cholesterol, Diabetes, Emphysema, Hypertension, MI, Other Heart Disease, Stroke

Formula:

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total_chronic_conditions = sum(chronic_vars == 1)
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5.2 Age Stratification

Continuous age variable was categorized into clinically meaningful groups:

1 (18–29 years): Young adults

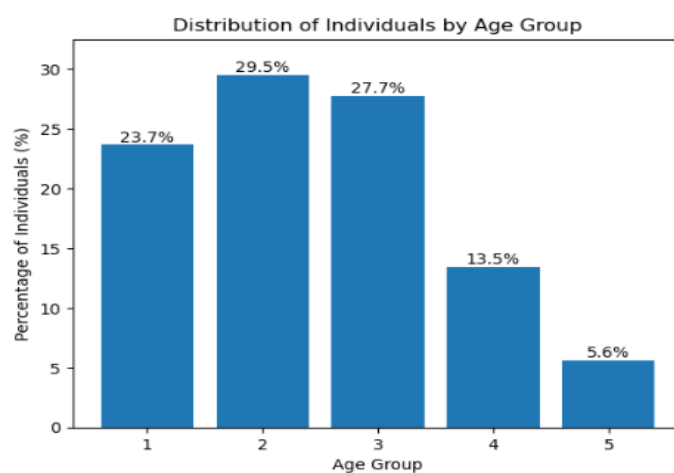
2 (30–44 years): Middle-aged adults

3 (45–59 years): Pre-elderly

4 (60–74 years): Young elderly

5 (75+ years): Elderly

Rationale: These categories align with Medicare eligibility (65+) and life stage healthcare utilization patterns.



5.3 BMI Categorization

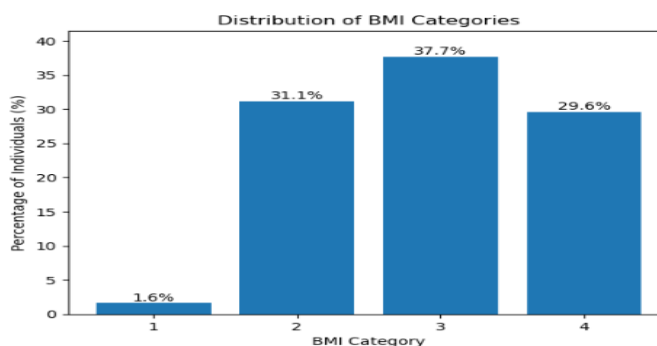
Body Mass Index was categorized using World Health Organization (WHO) standards:

1 (Underweight): BMI < 18.5

2 (Normal weight): $18.5 \leq \text{BMI} < 25.0$

3 (Overweight): $25.0 \leq \text{BMI} < 30.0$

4 (Obese): BMI ≥ 30.0



5.4 Binary Encoding of Yes/No Variables

Several variables were originally coded as 1 = “Yes” and 2 = “No.” Although binary in nature, retaining this coding may introduce unintended ordinal interpretation in machine learning models (i.e., assuming $2 > 1$).

To prevent this issue, the following variables were converted into proper binary indicators:

‘ADSMOK2’, ‘PHYACT3’, ‘JTPAIN1’, ‘VISION2’, ‘HEARNG2’, ‘EMPST1’, ‘TRIEVY1’, ‘MCREVY1’, ‘MCDEVY1’).

The recoding scheme was defined as:

1 (Yes) = 1 and 2 (No) = 0

This ensures that models treat these variables as categorical binary features rather than ordered numeric values, improving interpretability and preventing bias in estimation.

5.5 One-Hot Encoding

One-hot encoding was applied to nominal categorical variables to convert them into machine-readable format without introducing ordinal relationships.

The following variables were encoded:

'SEX', 'RACETHNX', 'MARRY1X', 'REGIONY1', 'INSCOVY1'.

This approach creates separate binary columns for each category, ensuring that models interpret these variables correctly as unordered categorical features.

5.6 Inflation Adjustment of Monetary Variables

Because the analysis combines MEPS panels from multiple years (2006–2011), all monetary variables were adjusted for inflation to ensure comparability across time. Healthcare expenditures reported in earlier years were converted to constant 2011 U.S. dollars using the Consumer Price Index (CPI).

Annual CPI values for 2006 through 2011 were obtained and used to construct an inflation adjustment factor.

The CPI for 2011 (224.9) was selected as the base year.

For each observation, an inflation multiplier was calculated as:

$$\text{Inflation Factor} = (\text{CPI}_{\text{year}} / \text{CPI}_{2011})$$

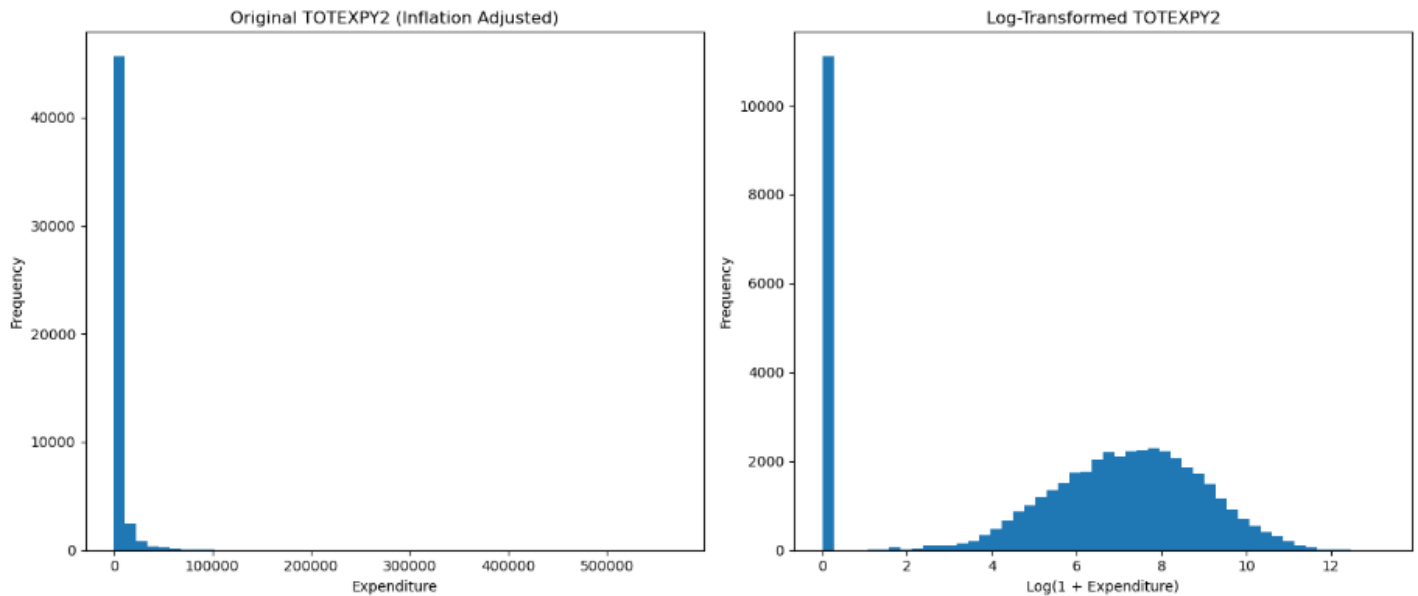
Each expenditure-related variable was then multiplied by its corresponding inflation factor to generate inflation-adjusted values. The following monetary variables were adjusted:

- Total expenditure (Year 1 and Year 2) - TOTEXPY2, TOTEXPY1,
- Self-paid expenditure - TOTSLFY1
- Medicaid expenditure - TOTMCDY1
- Private insurance expenditure - TOTPRVY1
- Public insurance expenditure - TOTOPUY1

For example:

$$\text{TOTEXPY2}_{\text{adjusted}} = \text{TOTEXPY2} \times \text{Inflation Factor}$$

5.7 Log Transformation and Expenditure Categorization



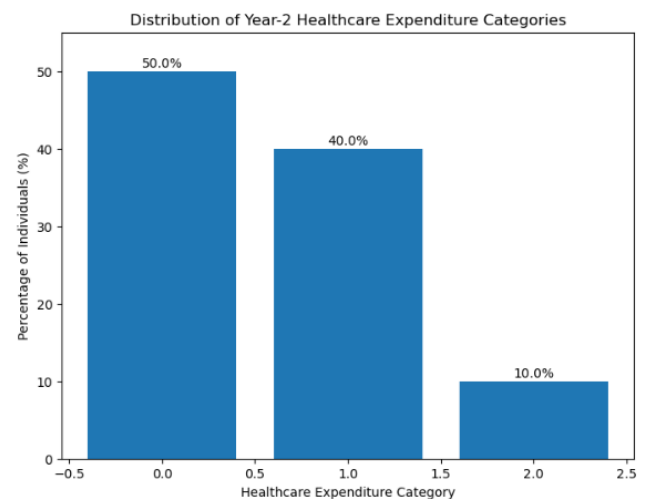
Healthcare expenditure data are typically highly right-skewed, with a small proportion of individuals incurring extremely high costs. To reduce skewness and stabilize variance, the inflation-adjusted Year-2 total expenditure variable (TOTEXPY2_adj) was log-transformed using the natural logarithm transformation:

$$\text{log_exp} = \log(1 + \text{TOTEXPY2_adj})$$

Initially, the categories were created using equal tertiles (33% each) to maintain perfectly balanced classes. However, this approach did not adequately capture the real-world structure of healthcare spending, where high-cost individuals represent a much smaller proportion of the population.

To better reflect the natural distribution of healthcare expenditures, the categorization strategy was revised to use unequal quantile thresholds, resulting in the following distribution:

Category	Description	Population(%)
0	Low Expenditure group	50%
1	Middle expenditure group	40%
2	high expenditure group	10%

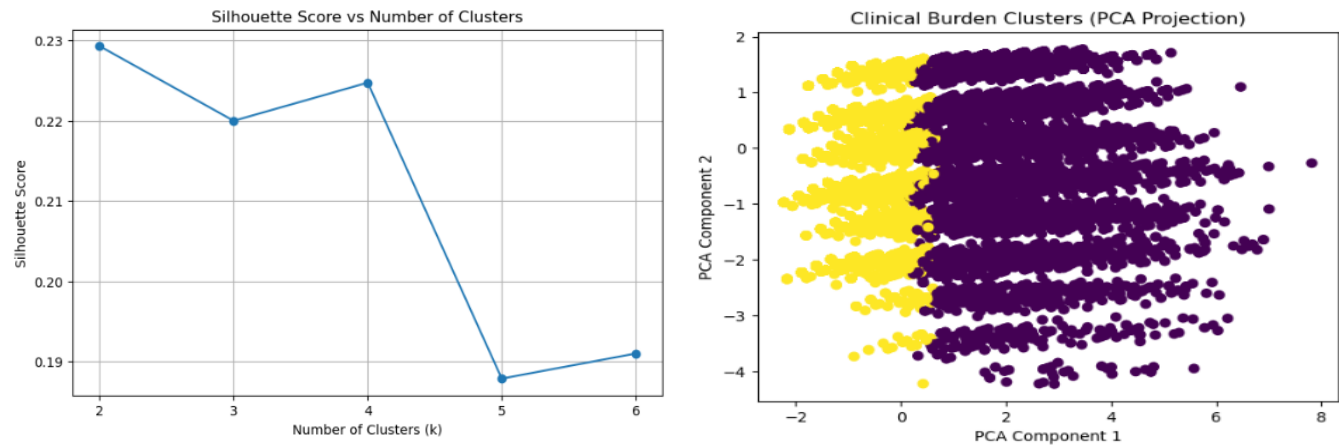


6 Patient Segmentation by Clinical Burden

6.1 Clustering Features

- Demographics ('age_group', 'SEX')
- Health factor ('JTPAIN1', 'VISION2', 'HEARNG2')
- Lifestyle (ADSMOK2, PHYACT3, bmi_category)
- Chronic burden (total_chronic_conditions)

6.2 Cluster Selection



The silhouette score plot indicates that k = 2 provide the optimal cluster separation, as it yields the highest score compared to other values of k.

The PCA projection shows two clearly distinguishable patient groups based on clinical burden, validating the clustering results with visible separation between low and high burden populations.

6.3 Cluster Profile:

Clinical Burden Cluster	Age Group	Sex	Joint Pain	Vision Impairment	Hearing Impairment	Smoking	Physical Activity	BMI Category	Total Chronic Conditions
0	3.48	1.57	61.0	0.14	0.17	19.0	44.0	3.21	2.67
1	2.01	1.52	7.0	0.02	0.01	17.0	62.0	2.83	0.43

Cluster	Size	Summary Description
0 (High Clinical Burden Group)	15,905 (34%)	Older age, Higher Chronic conditions, Higher Health Factor issues with less physical Activity.
1 (Low Clinical Burden Group)	33,915 (66%)	Younger age, Lower Chronic conditions, Lower Health Factor issues with High physical Activity.

This segmentation provides a strong foundation for:

- Risk-stratified predictive modeling
- Targeted intervention strategies

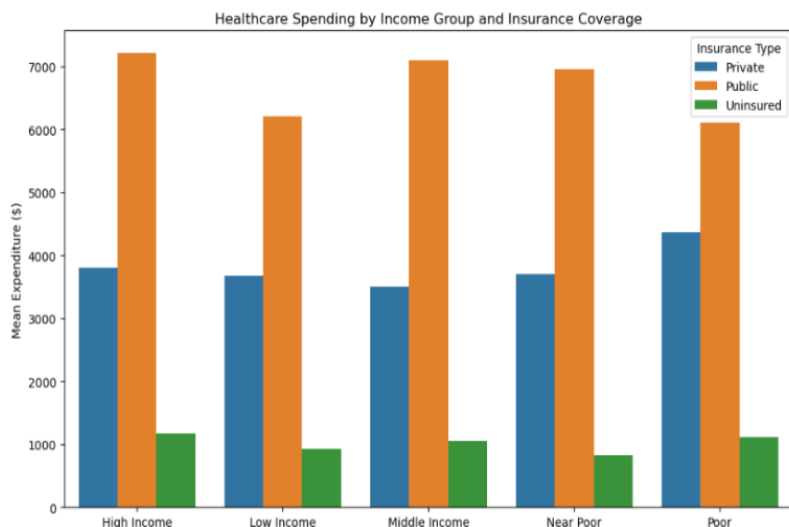
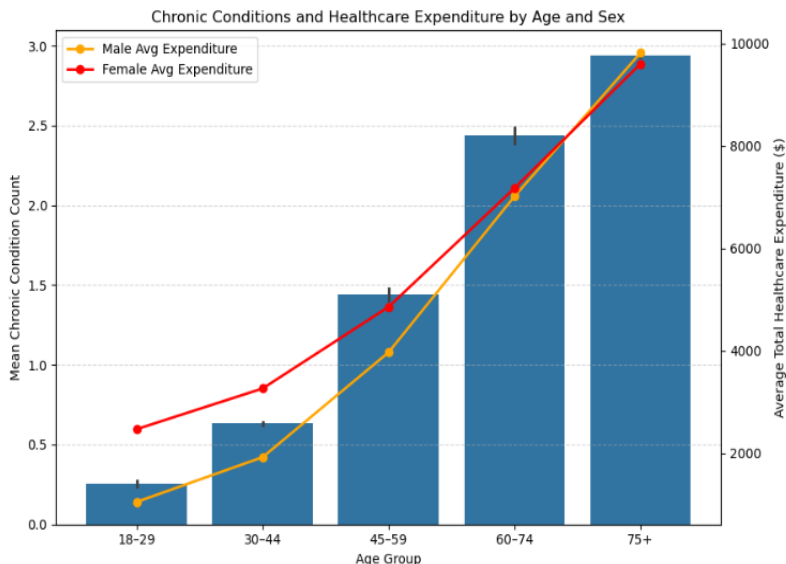
7 Exploratory Data Analysis (EDA)

Chronic Conditions and Healthcare Expenditure by Age and Sex:

Both chronic conditions and healthcare expenditure rise sharply with age, especially after 45.

Females spend slightly more than males in younger ages, but the gap closes in older groups (almost equal at 75+)

Strong positive relationship between chronic conditions and expenditure, more conditions = disproportionately higher cost



Healthcare Spending by Income Group and Insurance Coverage

Public insurance consistently has the highest spending across all income groups

Private spending is moderate and relatively stable, slightly higher for poorer groups

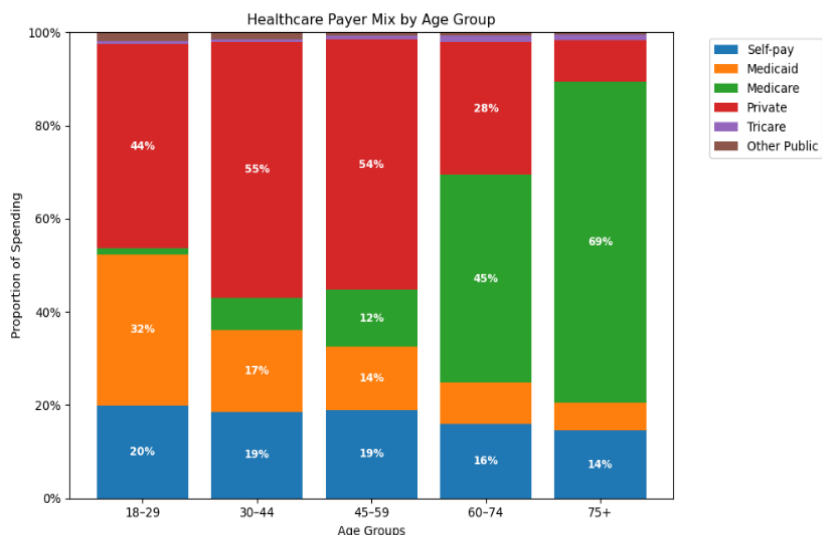
Uninsured individuals have the lowest spending across all groups.

Healthcare Payer Mix by Age Group

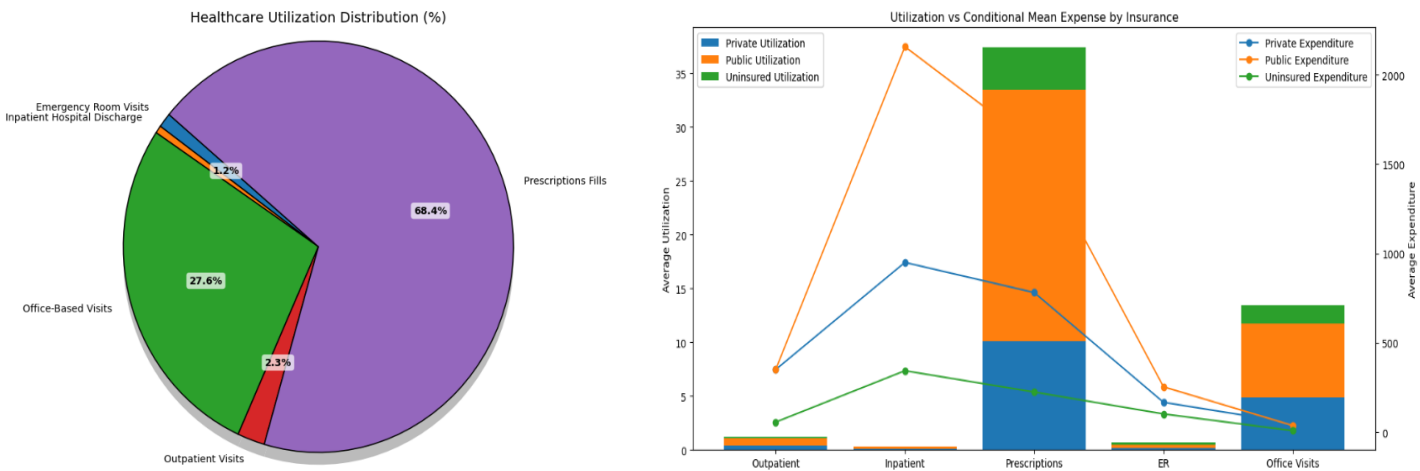
Medicare dominates spending in older age groups (60+), becoming the primary payer (~69% at 75+)

Private insurance is the main payer for younger and middle-aged groups, peaking around 30–59

Medicaid and self-pay proportions decline with age, showing a clear shift from mixed payers to Medicare-heavy coverage



Healthcare Utilization Distribution (%) & Spending by Insurance Type



Prescriptions dominate utilization (~68%) but not highest cost per service, while inpatient care is low volume but extremely high cost. Office visits are the second most common (27.6%) and moderate cost, making them a steady contributor to overall spending. Public insurance consistently shows highest spending per service, uninsured lowest.

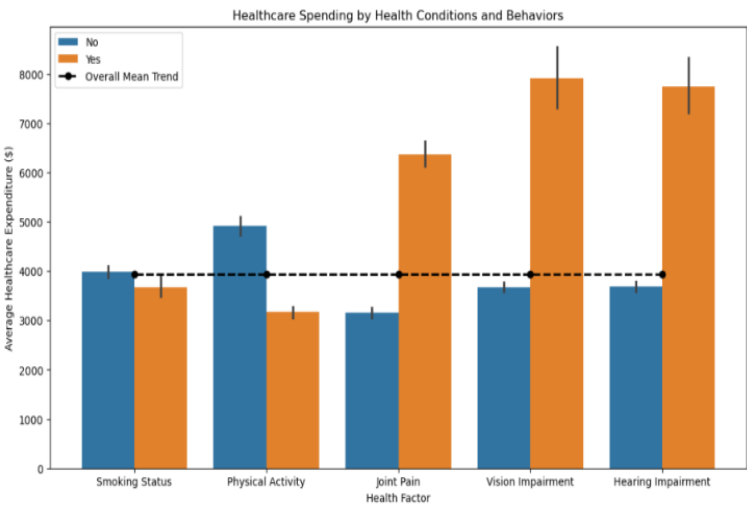


Healthcare Spending by BMI Category and Physical Activity

Obese category shows highest variability and extreme spending outliers, indicating strong link between high BMI and costly health events. Physically active individuals generally cluster at lower expenditure levels, especially in normal and overweight groups. Spending dispersion increases from normal to overweight to obese, suggesting risk and cost escalate non-linearly with BMI.

Healthcare Spending by Health Conditions and Behaviors

Health conditions (joint pain, vision, hearing issues) sharply increase spending, far above overall mean. Smoking and physical activity show mixed patterns, but not as impactful as other health conditions on expenditure. Behavioral factors alone don't spike costs as much as diagnosed conditions, highlighting that clinical burden drives major expenses.



8 Predictive Modeling

8.1 Objective of Predictive Modeling

The predictive modeling component aimed to classify individuals into three healthcare expenditure categories for Year2: Low expenditure, Medium expenditure, High expenditure

8.2 Modeling Framework

- **Target:**
TOTEXPY2_category
- **Predictors:**
 - Demographic** (REGIONY1, AGEY1X, SEX, RACETHNX, MARRYY1X) +
 - Socioeconomic** ('POVCATY1', 'employed_flag',) +
 - Health Factors** ('RTHLTH1', 'MNHLTH1', 'VISION2', 'HEARNG2', 'JTPAIN1', 'bmi_category', 'total_chronic_conditions', 'clinical_burden_cluster') +
 - Lifestyle factors** ('PHYACT3', 'ADSMOK2') +
 - Insurance Coverage** ('INSCOVY1', 'TRIEVY1', 'MCREVY1', 'MCDEVY1', 'TOTSLFY1_adj', 'TOTMCDY1_adj', 'TOTMCRY1_adj', 'TOTPRVY1_adj', 'TOTTRIY1_adj', 'TOTOPUY1_adj') +
 - Healthcare utilization** ('OBTOTVY1', 'OPTOTVY1', 'ERTOTY1', 'IPDISY1', 'RXTOTY1')+
 - Total Healthcare Expenditure** ('TOTEXPY1_adj')
- **Train Test Dataset split (70:30):**
To evaluate predictive performance, the dataset was divided into training and testing sets using a 70:30 split, ensuring that 70% of observations were used for model training and 30% for out-of-sample evaluation. Stratified sampling was applied to preserve balanced distribution across expenditure categories.

8.3 Model Performance Summary

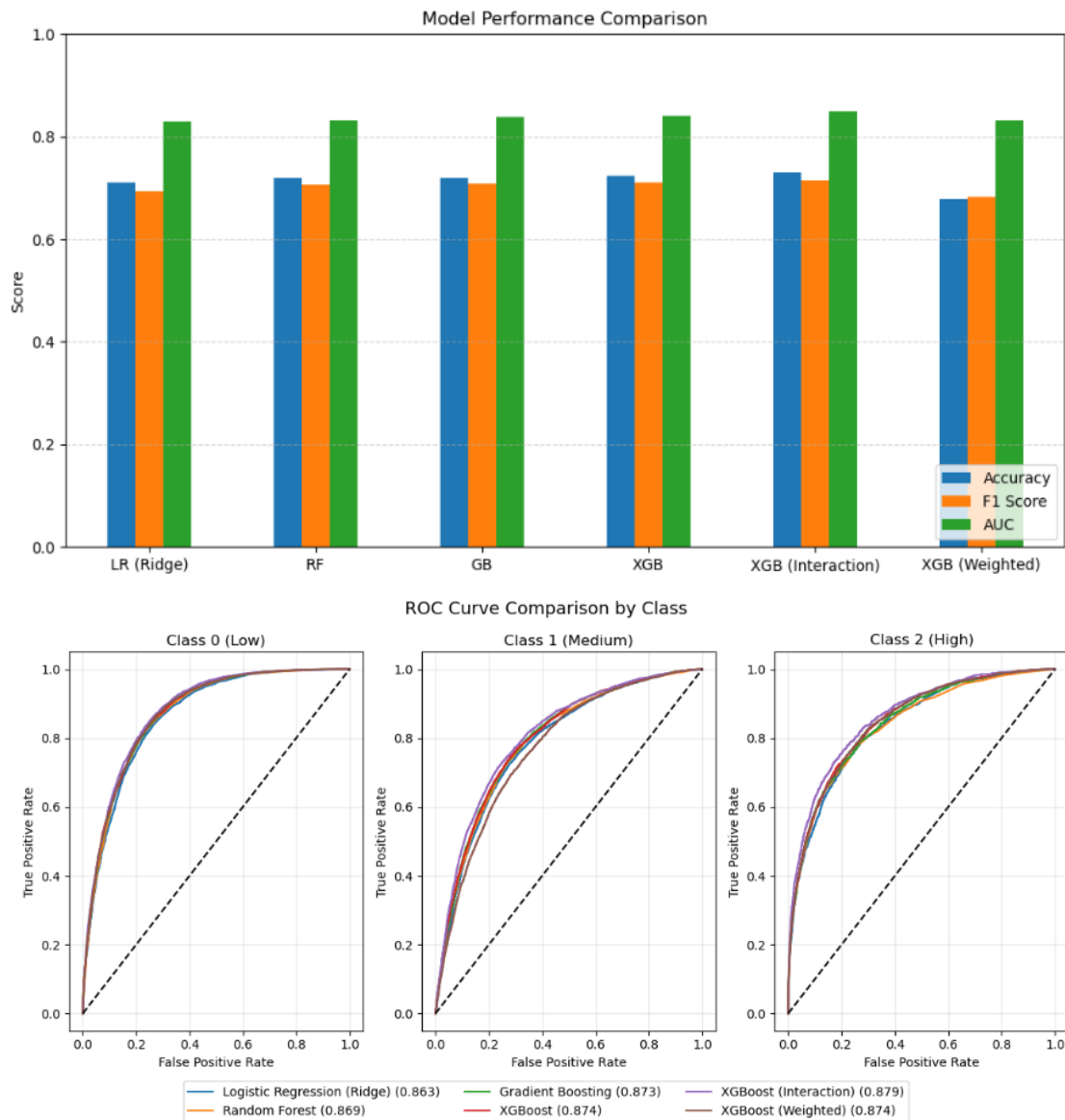
Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1 Score	Macro F1 Score
Logistic Regression	0.71	0.71	0.71	0.69	0.59
Random Forest	0.72	0.72	0.72	0.71	0.62
XGBoost	0.72	0.72	0.72	0.71	0.62
Ordinal Model	0.71	0.71	0.71	0.70	0.61
Interaction Model	0.71	0.71	0.71	0.70	0.62
XGBoost (Weighted)	0.68	0.69	0.68	0.68	0.62

8.4 Class-wise Model Performance (F1 Score)

Model	Class 0 (Low Expenditure)	Class 1 (Medium Expenditure)	Class 2 (High Expenditure)
Logistic Regression	0.80	0.65	0.32
Random Forest	0.80	0.67	0.38
XGBoost	0.80	0.68	0.38
Ordinal Model	0.80	0.67	0.38
Interaction Model	0.79	0.66	0.41
XGBoost (Weighted)	0.80	0.59	0.46

The F1-score for the high-cost class detection is improved by approximately 44%, from 0.32 (Baseline model) to 0.46, indicating a substantial gain in detecting the most critical patient group.

8.5 Model Selection and Justification



The final model was selected based on its ability to effectively identify high-cost individuals, rather than solely maximizing overall accuracy. Since high-cost patients represent a small but critical group, improving their detection was prioritized.

- Standard models (Random Forest, XGBoost) achieved higher accuracy (~72%). However, they showed limited performance for high-cost class (F1 $\approx$ 0.38)
- Weighted XGBoost improved high-cost detection to F1 = 0.46 (~44% improvement)
- Trade-off observed:
 - Slight drop in overall accuracy (72% to 68%)
 - Reduced performance for medium class
- Class weighting helped address class imbalance (10% high-cost group)

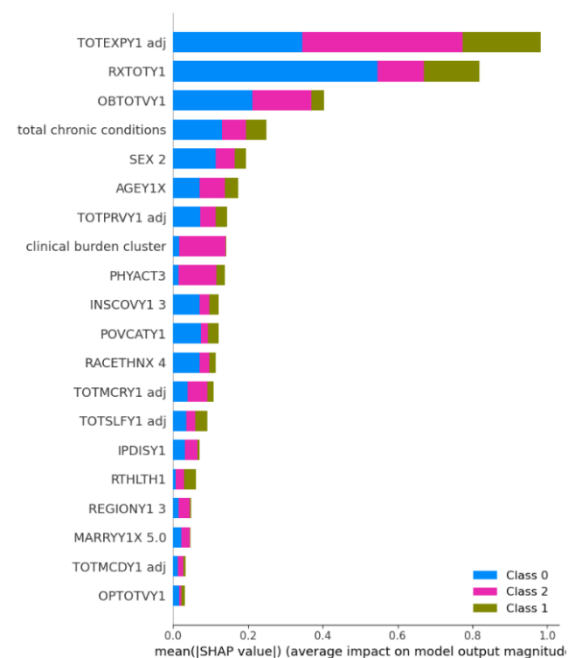
Final Decision

Although the standard XGBoost model achieved the highest overall accuracy (~72%), it underperformed in identifying high-cost patients, which represent the most critical segment in healthcare expenditure analysis. To address this limitation, a class-weighted XGBoost model was implemented to account for class imbalance. While this resulted in a reduction in overall accuracy (~68%), it significantly improved the F1-score for the high-cost class from 0.32 to 0.46. Given the objective of early identification of high-expenditure individuals for targeted intervention, the weighted XGBoost model was selected as the final model, as it better aligns with the practical and policy-driven goals of the study.

- Weighted XGBoost selected as final model
- Better aligned with real-world objective: identifying high-risk patients
- Prioritizes impact over overall accuracy

8.6 Feature Importance:

- Prior expenditure (TOTEXPY1_adj) is the strongest predictor, showing that past spending heavily influences future healthcare costs.
- Healthcare utilization features (RXTOTY1, OBTOTVY1) are key drivers, indicating frequent service use leads to higher future expenditure.
- Chronic disease burden (total_chronic_conditions) significantly impacts predictions, increasing the likelihood of high-cost classification.
- Demographic factors (age, sex) have moderate influence, with older individuals contributing more to higher cost categories.
- Clinical indicators (joint pain, vision, hearing issues) help identify individuals with elevated healthcare needs.
- Insurance and socioeconomic variables have relatively lower impact compared to utilization and clinical features.
- Overall, the model relies more on historical spending and utilization patterns than demographic or socioeconomic factors.



9 Limitations

- Moderate performance for high-cost class (F1 = 0.46)
- Healthcare expenditure is highly skewed and difficult to predict
- MEPS data is self-reported, introducing potential bias
- Model uses Year-1 features only, missing temporal dynamics
- Class imbalance leads to trade-offs in model performance

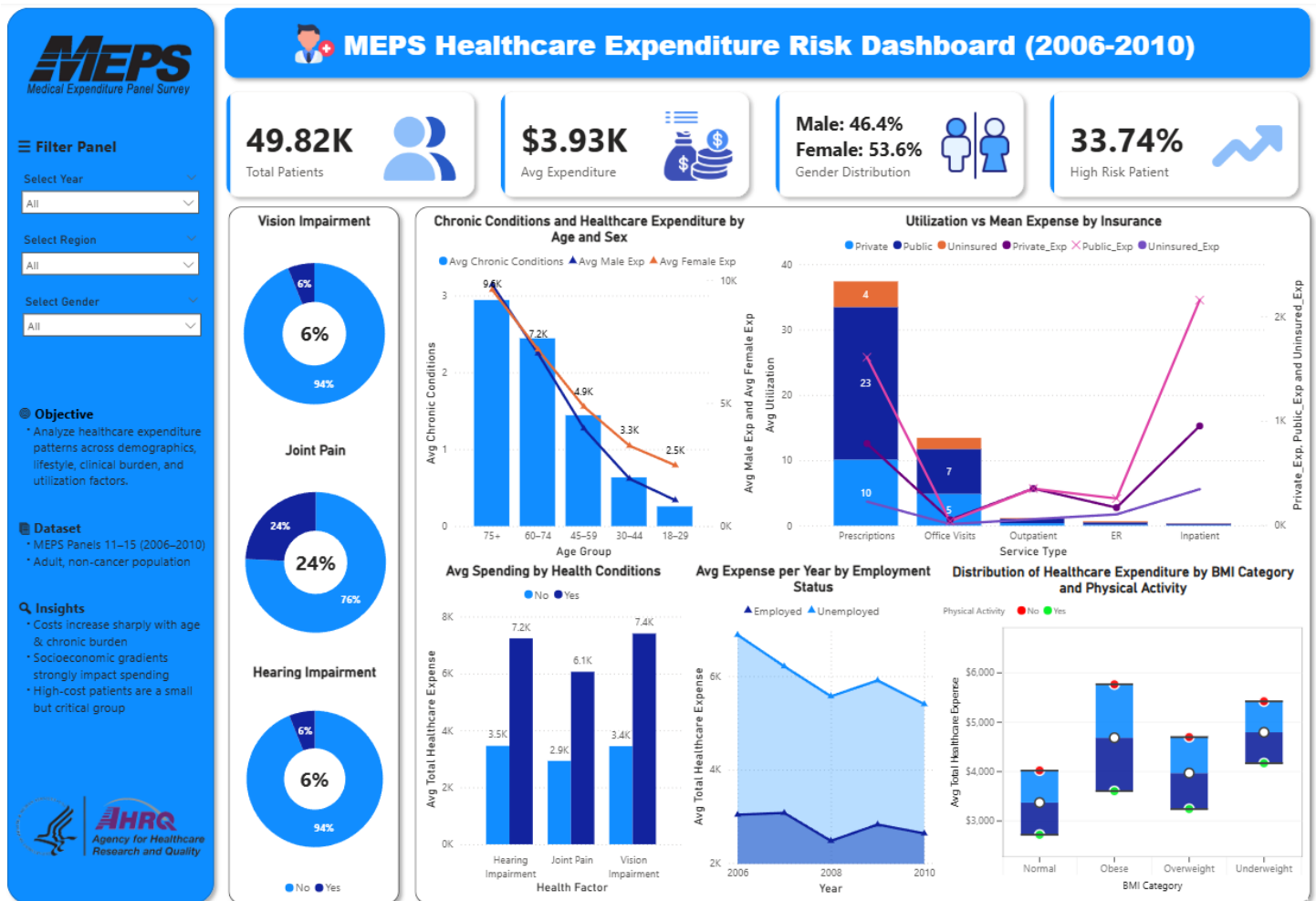
10 Future Work

- Apply SMOTE / advanced resampling to improve minority class detection
- Use two-part (hurdle) models for better cost modeling
- Implement SHAP analysis for model interpretability
- Experiment with cost-sensitive learning and hybrid models

11 Conclusion

This study successfully developed a longitudinal predictive modeling framework to classify healthcare expenditure levels among non-cancer adults using MEPS data. Through rigorous data preprocessing, feature engineering, and exploratory analysis, the study identified key drivers of healthcare costs, including prior expenditure, healthcare utilization, and chronic disease burden. While multiple machine learning models demonstrated similar overall performance, the weighted XGBoost model was selected as the final model due to its superior ability to detect high-cost individuals, which are the most critical segment for healthcare planning and intervention. Despite a slight reduction in overall accuracy, the model achieved a substantial improvement in high-cost prediction, highlighting the importance of prioritizing real-world impact over aggregate metrics. Overall, the study provides a robust foundation for healthcare risk stratification and offers valuable insights for targeted policy and resource allocation strategies.

12 Final Power BI Dashboard



13 Project Key Insights:

EDA Insights:

- Healthcare costs are highly skewed (few patients drive most spending)
- Costs increase with age and chronic conditions
- Prior utilization is strongest predictor of future expenditure
- Public insurance shows highest spending patterns, uninsured the lowest.
- Clinical conditions (joint pain, hearing/vision issue) drive cost more than lifestyle factors (smoking)
- Obese and Underweight category shows higher spending than normal and overweight category.
- Unemployed people spend more than employed ones.
- Medicare dominates spending in older age groups (60+), Private insurance is the main payer for younger and middle-aged groups (30-60)
- Identified 2 distinct clusters, High-burden group (older, more conditions, less active) and Low-burden group (younger, healthier, more active) which enables risk-based patient stratification.

Predictive Modeling:

- Models achieved ~71–72% accuracy overall
- High-cost patients are hardest to predict (imbalanced data)
- Baseline high-cost detection improved 0.32 → 0.46 (~44%)
- Weighted XGBoost is best for high-cost identification
- Trade-off: lower accuracy but better real-world impact
- Model driven by prior cost, utilization, and chronic burden

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